PRAFUL PATEL
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Professional Summary:

- 7 years of experience in Embedded Systems development, primarily in the automotive domain.
- Strong working knowledge of C, C++, Standard Template Library (STL), and Python for embedded and application-level development.
- Proficient in Embedded C/C++ programming, with a solid understanding of low-level system design and hardware-software interaction.
- Hands-on experience with Linux-based embedded platforms, focused on user-space application development, process management, and system-level debugging.
- Adept at working in Agile development teams with a strong focus on software quality, MISRA compliance, and unit testing.

Professional Experience:

T-Systems ICT India Pvt ltd. – June 2022 to Present.

Tools & IDEs: Polarion, VS Code, Git/GitLab, Jenkins, GTest, EA, Confluence, Cppcheck, clang-tidy, Docker, make, cmake, valgrind, ThreadSanitizer, JIRA

Standards & Protocols: ISO 26262, MISRA C++, MQTT, TUFT, TCP/IP, UDS, CAN

AI & Libraries: ChatGPT, GitHub Copilot, cURL, ZipLib, OpenSSL, XmlRpc, CJSON, paho-mqtt

Responsibilities and project details:

Project name: OTA (Over the air) Software

Language: C++, C

Role: Senior Software Engineer

Project description:

Developed an in-car software component for OTA (Over-the-Air) updates, enabling ECU software upgrades via a master ECU interface. The OTA software manages download and distribution of update packages from the backend server to target ECUs, coordinating with low-level flashing software for installation.

Responsibilities:

- Hands-on experience in design, development, and testing of embedded software systems.
- Proficient in implementing Unified Diagnostic Services (UDS) and CAN socket communication protocols.
- Strong expertise in developing and debugging multithreaded applications, including thread synchronization, inter-process communication, and performance optimization.
- Steering Agile-Scrum meetings and assisting Project Manager in planning, deciding timelines in sprint meetings.
- Conducted software reviews, root cause analysis, effort estimation, task planning, and ensured process adherence for project quality and efficiency.
- Proficient in debugging embedded code to identify and resolve issues promptly.

Capgemini Technology Services India Ltd (**Client**: Robert Bosch Engineering and Business Solution Private Ltd, Bangalore) – October 2018 to June 2022.

IDE and tools: Eclipse, CLion, Trace32, IBM DOORS, Jira, Git, ClearQuest, Cantata, Google Test (GTest)

Standards & Protocols: ISO 26262, MISRA C/C++, AUTOSAR C++14, CAN, LIN, SPI, I^2C, UART**,** UDS

Language: C, C++

Role: Software Engineer

Project name: Parking Assistance (PA)

Responsibilities:

- Developed and maintained parking control functionalities in an embedded automotive environment.
- Modified and optimized existing code to support new customer requirements, ensuring system reliability and performance.
- Responsible for bug fixing and implementing change requests throughout the development lifecycle.
- Hands-on experience on various tools like Configuration Management Tool (GIT), Requirement Management Tool (DOORS).
- Performed unit and module testing using Google Test (GTest) to ensure code quality and correctness.

Project name: High-Definition Map (HDMAP)

Responsibilities:

- Developed and optimized map-matching algorithms for high-precision localization in autonomous driving systems.
- Implemented code modifications to support evolving customer requirements and performed detailed debugging and validation.
- Collaborated directly with customers and cross-functional teams to clarify technical requirements and provide ongoing support.
- Led and participated in software reviews, performed root cause analysis for critical issues, and contributed to effort estimation and task planning.
- Ensured adherence to quality processes and standards throughout the software development lifecycle.

Achievement:

- Contributed to the successful Automotive SPICE (ASPICE) Capability Level 2 assessment by actively participating in process areas SWE.3, SWE.4, and SWE.5.
- Completed Advanced C++ certification course on Udemy, enhancing proficiency in modern C++ concepts including STL, OOP, and templates.
- Completed Python Programming certification course on Coursera, gaining strong foundational skills in scripting, data structures, and automation.

Education:

- B.Tech in Electronics and Communication Engineering (ECE), Dharmsinh Desai University, Nadiad, Gujarat — September 2013 to June 2016.
- Diploma in Electronics and Communication Engineering (ECE), Government Polytechnic**,** Ahmedabad**,** Gujarat — June 2010 - May 2013.